



Name: _____

Teacher: _____

Class: _____

FORT STREET HIGH SCHOOL

2009

PRELIMINARY SCHOOL CERTIFICATE COURSE

ASSESSMENT TASK 3: FINAL PRELIMINARY EXAMINATION

Mathematics Extension I

TIME ALLOWED: 1½ HOURS

Outcomes Assessed	Questions	Marks
Demonstrates appropriate mathematical techniques algebra, equations and graphs.	1	
Manipulates algebraic expressions to solve problems from the topic areas of calculus and circle geometry.	2	
Applies appropriate techniques to solve problems from the topic areas of co-ordinate geometry, plane geometry and 3D trigonometry.	3	
Applies appropriate techniques to solve problems in trigonometry.	4	
Applies appropriate techniques to solve problems in quadratics, locus and polynomials.	5	

Question	1	2	3	4	5	Total	%
Marks	/17	/13	/15	/15	/15	/75	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started in a new booklet

2009 Final Preliminary Exam

Question 1 (17 marks)

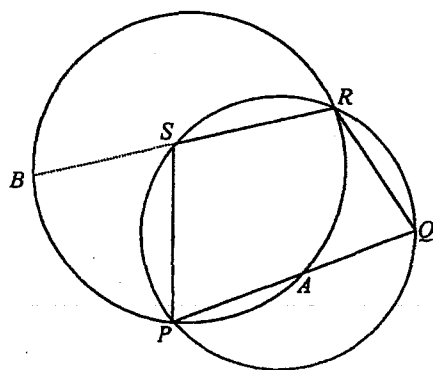
- | | <i>marks</i> |
|---|--------------|
| a. Factorise $3x^2 - 5x - 2$ | 1 |
| b. Solve $ 2x - 3 > 5$ | 2 |
| c. Find the values of a and b given that $(ax - 3)^2 + b \equiv 4x^2 - 12x + 15$ | 2 |
| d. A weight oscillates up and down at the end of an elastic string. Its depth below the point of suspension, t seconds after it starts, is $(12 + 3 \cos 2\pi t)$ cm. At what time will its depth be 9cm? | 2 |
| e. | 3 |
| i. What is the period of $y = 3 + \sin 2x$ | |
| ii. What is the amplitude of $y = 3 + \sin 2x$ | |
| iii. Draw a neat sketch of $y = 3 + \sin 2x$ where $0 \leq x \leq 2\pi$ | |
| f. Draw a sketch of the function $f(x) = \frac{1}{x^2 - 1}$, stating the domain and range | 4 |
| g. Solve $\frac{4x}{x+2} \leq 1$ | 3 |

Question 2 (13 marks)

- | | <i>marks</i> |
|---|--------------|
| a. The interval AB has end points $A(-4, 1)$ and $B(8, 7)$. Find the coordinates of the point P that divides the interval AB internally in the ratio 1 : 3 | 2 |
| b. Differentiate with respect to x | 8 |
| i. $y = \sqrt{x^2 + 5}$ | |
| ii. $y = \frac{1}{2x^3}$ | |
| iii. $y = 2x\sqrt{x}$ | |
| iv. $y = \frac{x-3}{\sqrt{x}}$ | |

c.

3



A circle through the points P , A and R cuts RS produced at B . Prove that $AB \parallel SQ$.

Question 3 (15 marks)

marks

a. $A(2,2)$, $B(-2,-1)$, $C(2,-1)$ are the vertices of a triangle and E is the points $(1,0)$

i. Show that the equation of AB is $3x - 4y + 2 = 0$

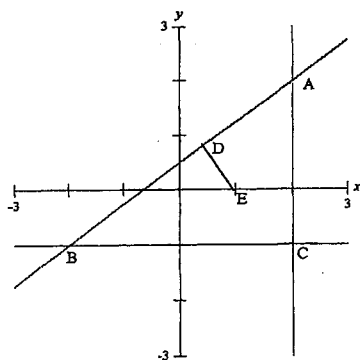
2

ii. D is the foot of the perpendicular from $E(1,0)$ to AB . Find the equation of ED

2

iii. Find the coordinates of D .

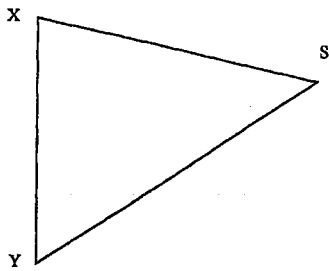
2



b. Use the k method to find the equation of the straight line through the point $(-4,-1)$ that passes through the intersection of the line $2x + y - 1 = 0$ and $3x + 5y + 16 = 0$.

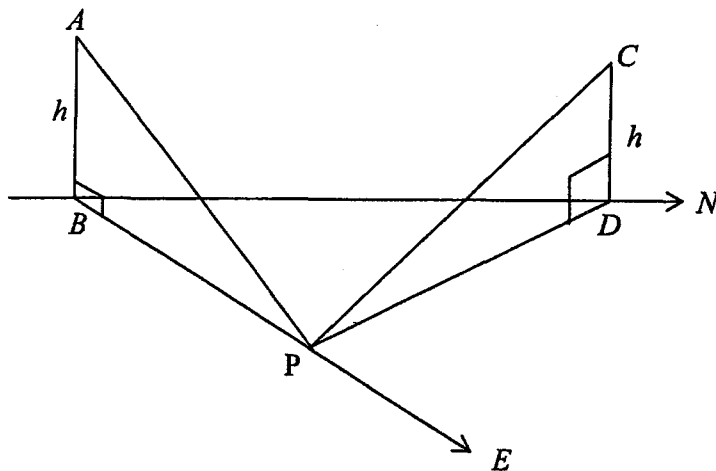
2

- c. Two lighthouses X and Y , 12 kilometres apart observe a ship S , at the same instant. X and Y are on a north-south line. From X , the ship is on a bearing of 105° and from Y the bearing of the ship is 058°



Copy the diagram onto your own paper.

- i. Show that $\angle XSY = 47^\circ$ 1
 - ii. Find, correct to one decimal place the distance of the ship from the nearer lighthouse. 2
- d. AB and CD are towers of equal height (h). CD is due north of AB . From a point P on the same horizontal plane as the feet B and D of the towers, and bearing due east of the tower AB , the angles of elevation of A and C , the tops of the towers, are 47° and 31° respectively. If the distance between the towers is 88m , find the height of the tower to the nearest metre. 4



Question 4 (15 marks)

marks

a. Solve $\cos 2\alpha = \frac{1}{2}$, where $0 \leq \alpha \leq 2\pi$

3

b.

i. Prove that $\frac{\sin 2x}{1 - \cos 2x} = \cot x$

3

ii. Hence, or otherwise, obtain a value for $\cot 67\frac{1}{2}^\circ$

3

(Rationalise the denominator)

c. Find the general solution of the equation $\cos\left(x - \frac{\pi}{4}\right) = -\frac{1}{2}$.

2

d. Express $7 \cos \theta - \sin \theta$ in the form $R \cos(\theta + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq 360$.

4

Hence solve the equation $7 \cos \theta - \sin \theta = 5$ for $0 \leq \theta \leq 360$

Question 5 (15 marks)

marks

a. Find the coordinates of the vertex and the focus, and the equation of the directrix of the parabola with equation $x^2 - 10x - 16y - 7 = 0$

5

b. Factor and sketch the polynomial $P(x) = x^4 - x^3 - 19x^2 - 11x + 30$, indicating all intercepts with the axes.

5

c. If α, β, γ are the roots of the equation $2x^3 + 5x - 3 = 0$, find the values of

i. $\alpha + \beta + \gamma$

1

ii. $\alpha^2 + \beta^2 + \gamma^2$

2

iii. $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$

2

Year 11 Extension 1 Assessment 3 Solutions

Q1

a. $(3x+1)(x-2)$

b. $|2x-3| > 5$

$$2x-3 > 5 \quad - (2x-3) > 5$$

$$2x > 8 \quad 2x-3 < -5$$

$$x > 4 \quad 2x < -2$$

$$x < -1$$

c.

$$(ax-3)^2 \equiv 4x^2 - 12x + 15$$

$$a^2x^2 - 6ax + 9 + b \equiv 4x^2 - 12x + 15$$

$$a = 2$$

$$9 + b = 15$$

$$b = 6$$

d.

$$9 = 12 + 3\cos 2\pi t$$

$$-3 = 3\cos 2\pi t$$

$$-1 = \cos 2\pi t$$

$$\pi = 2\pi t$$

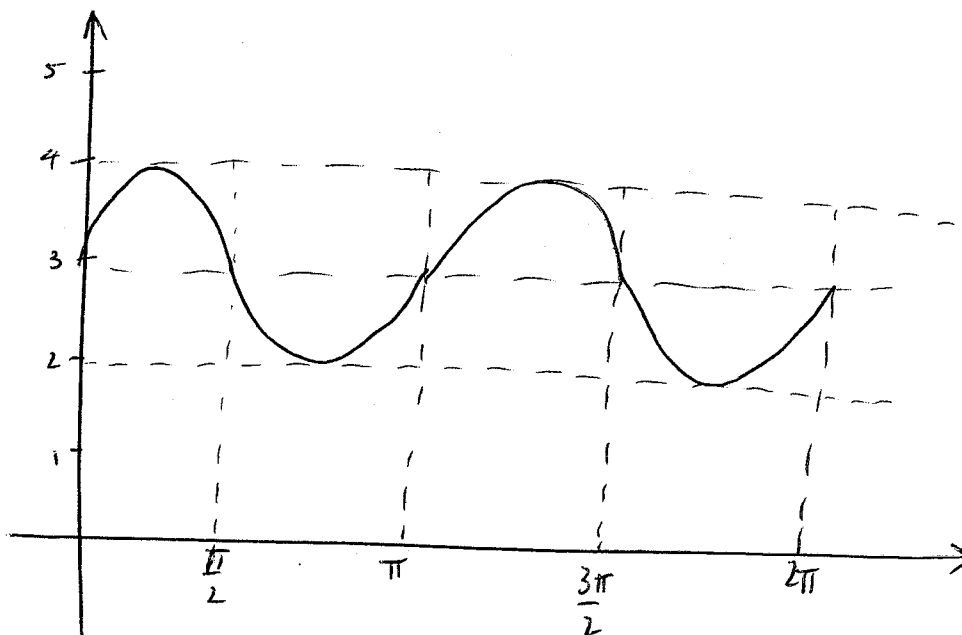
$$t = \frac{1}{2}s$$

e.

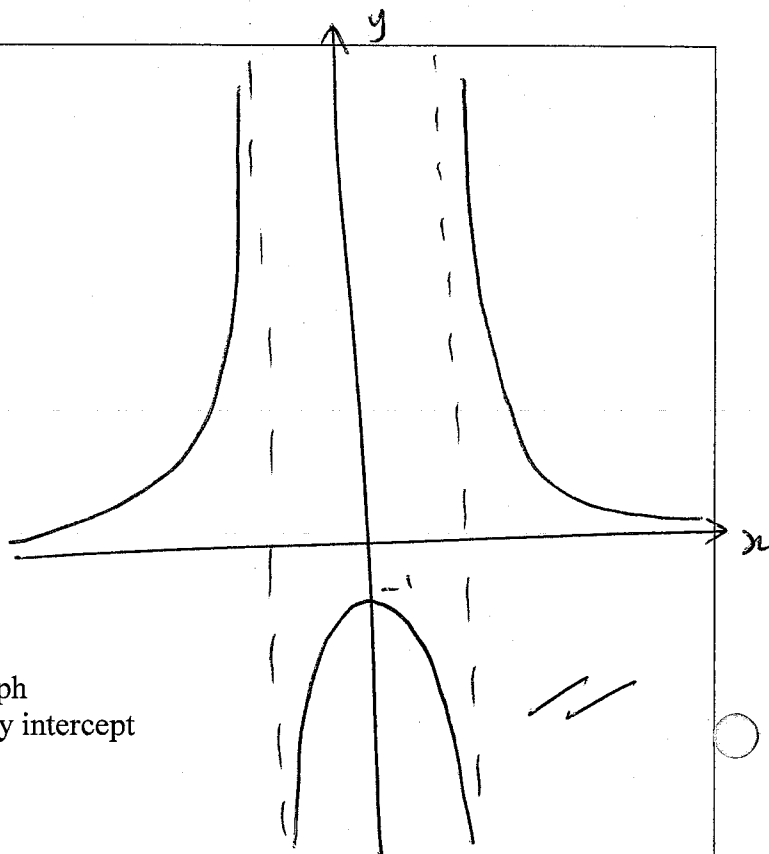
$$T = \pi$$

$$a = 1$$

c.



f.



1 mark for the correct shape of the graph
1 mark for the correct asymptotes and y intercept

Domain: All real x , $x \neq \pm 1$

Range: $-1 \geq y > 0$

g.

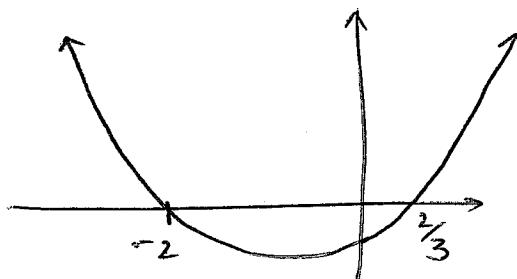
$$4x(x+2) \leq (x+2)^2$$

$$0 \leq (x+2)^2 - 4x(x+2)$$

$$0 \leq (x+2)(-3x+2)$$

$$(x+2)(3x-2) \leq 0$$

$$-2 < x \leq \frac{2}{3}$$



Q2

a.

$$x = \frac{mx_2 + nx_1}{m+n}$$

$$y = \frac{my_2 + ny_1}{m+n}$$

$$x = \frac{1(8) + 3(-4)}{4}$$

$$y = \frac{1(7) + 3(1)}{4}$$

$$x = \frac{-4}{4} \quad \checkmark$$

$$y = \frac{10}{4} \quad \checkmark$$

$$(-1, 2\frac{1}{2})$$

b.

$$\text{i. } y' = \frac{1}{2}(x^2 + 5)^{-\frac{1}{2}} \cdot 2x = \frac{x}{\sqrt{x^2 + 5}} \quad \checkmark$$

ii.

$$y = \frac{1}{2}x^{-3} \quad \checkmark$$

$$y' = -\frac{3}{2}x^{-4} = -\frac{3}{2x^4} \quad \checkmark$$

iii.

$$y = 2x\sqrt{x} = 2x^{\frac{3}{2}} \quad \checkmark$$

$$y' = 2 \cdot \frac{3}{2} x^{\frac{1}{2}} = 3\sqrt{x} \quad \checkmark$$

iv.

$$y' = \frac{\sqrt{x} \cdot 1 - (x-3) \frac{1}{2} x^{-\frac{1}{2}}}{x} \quad \checkmark$$

$$= \frac{\sqrt{x} - \frac{x-3}{2\sqrt{x}}}{x}$$

$$= \frac{x+3}{2x\sqrt{x}} \quad \checkmark$$

c.

$\angle BRP = \angle BAP$ (angles at the circ. subtended by the same arc) $\checkmark$

$\angle SRP = \angle SQA$ (angles at the circ. subtended by the same arc) $\checkmark$

$\angle BAP = \angle SQA$

$\angle BAP$ AND $\angle SQA$ are corresponding angles $\checkmark$

$\therefore BA \parallel SQ$

Q3

a. P

i. $m = \frac{-1-2}{-2-2} = \frac{3}{4} \quad \checkmark$

$$y-2 = \frac{3}{4}(x-2)$$

$$4y-8 = 3x-6 \quad \checkmark$$

$$0 = 3x - 4y + 2 \dots\dots\dots(1)$$

ii.

$$m = -\frac{4}{3} \quad \checkmark$$

$$y-0 = -\frac{4}{3}(x-1) \quad \checkmark$$

$$4x+3y-4 = 0 \dots\dots\dots(2)$$

iii.

$$(1) \times 4 - (2) \times -3$$

$$12x - 16y + 8 - 12x - 9y + 12 = 0$$

$$-25y + 20 = 0$$

$$y = \frac{20}{25} = \frac{4}{5}$$

$$4x + 3\left(\frac{4}{5}\right) - 4 = 0$$

$$4x = 4 - \frac{12}{5}$$

$$4x = \frac{8}{5}$$

$$x = \frac{8}{20} = \frac{2}{5}$$

b. $(2x + y - 1) + k(3x + 5y + 16) = 0$

at $(-4, -1)$

$$(2(-4) - 1 - 1) + k(3(-4) + 5(-1) + 16) = 0$$

$$-10 - k = 0$$

$$k = -10$$

$$(2x + y - 1) - 10(3x + 5y + 16) = 0$$

$$2x + y - 1 - 30x - 50y - 160 = 0$$

$$-28x - 49y - 161 = 0$$

$$4x + 7y + 23 = 0$$

c.

i. $\angle YXS = 75^\circ$ (supplementary angles)

$$\angle XSY = 180 - 75 - 58 = 47^\circ \text{ (angle sum of a triangle)}$$

ii.

$$\frac{12}{\sin 47} = \frac{SY}{\sin 75}$$

$$SY = \frac{12}{\sin 47} \times \sin 75$$

$$SY = 15.85$$

$$\frac{12}{\sin 47} = \frac{SX}{\sin 58}$$

$$SX = \frac{12}{\sin 47} \times \sin 58$$

$$SX = 13.91$$

OR 1 mark for correct answer and a mark for a reason

d.

$$\tan 47 = \frac{h}{PB} \quad \tan 31 = \frac{h}{PD}$$

$$PB = \frac{h}{\tan 47} \quad PD = \frac{h}{\tan 31}$$

$$88^2 + PB^2 = PD^2$$

$$88^2 = \left(\frac{h}{\tan 31} \right)^2 - \left(\frac{h}{\tan 47} \right)^2$$

$$88^2 = h^2 \left(\frac{1}{\tan^2 31} - \frac{1}{\tan^2 47} \right)$$

$$88^2 = h^2 (\cot^2 31 - \cot^2 47)$$

$$h^2 = \frac{88^2}{\tan^2 59 - \tan^2 43}$$

$$h = 64$$

Q4

a.

$$\cos 2\alpha = \frac{1}{2} \quad 0 \leq \alpha \leq 2\pi$$

$$0 \leq 2\alpha \leq 4\pi$$

$$2\alpha = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$\alpha = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

b.

i.

$$LHS = \frac{\sin 2x}{1 - \cos 2x}$$

$$= \frac{2 \sin x \cos x}{1 - 2 \cos^2 x + 1}$$

$$= \frac{2 \sin x \cos x}{2 - 2 \cos^2 x}$$

$$= \frac{2 \sin x \cos x}{2 \sin^2 x}$$

$$= \frac{\cos x}{\sin x}$$

$$= \cot x = RHS$$

ii.

$$\begin{aligned}
 \cot 67\frac{1}{2}^\circ &= \frac{\sin 2\left(67\frac{1}{2}\right)}{1 - \cos\left(67\frac{1}{2}\right)} \\
 &= \frac{\sin 135}{1 - \cos 135} \\
 &= \frac{1}{1 + \frac{1}{\sqrt{2}}} \\
 &= \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2} + 1} \\
 &= \frac{1}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1} \\
 &= \sqrt{2} - 1
 \end{aligned}$$

c.

$$\begin{aligned}
 \cos\left(x - \frac{\pi}{4}\right) &= -\frac{1}{2} \\
 \left(x - \frac{\pi}{4}\right) &= 2n\pi \pm \frac{2\pi}{3} \\
 x &= 2n\pi \pm \frac{2\pi}{3} + \frac{\pi}{4} \\
 x &= 2n\pi + \frac{11\pi}{12}, 2n\pi - \frac{5\pi}{12}
 \end{aligned}$$

d.

$$\begin{aligned}
 R &= \sqrt{7^2 + 1^2} = \sqrt{50} = 5\sqrt{2} \\
 \tan \alpha &= \frac{1}{7} = 8.1^\circ \\
 5\sqrt{2} \cos(x + 8.1) &= 5 \\
 \cos(x + 8.1) &= \frac{1}{\sqrt{2}} \\
 x + 8.1 &= 45 \\
 x &= 36.9^\circ
 \end{aligned}$$

Q5

a.

$$\begin{aligned}
 x^2 - 10x - 16y - 7 &= 0 \\
 x^2 - 10x + 25 &= 16y + 7 + 25 \\
 (x - 5)^2 &= 16y + 32 \\
 (x - 5)^2 &= 16(y + 2)
 \end{aligned}$$

vertex (5, -2)

$a = 4$

focus (5, 2)

directrix $y = -6$

b.

$$P(x) = x^4 - x^3 - 19x^2 - 11x + 30$$

$$P(1) = 1 - 1 - 19 - 11 + 30 = 0$$

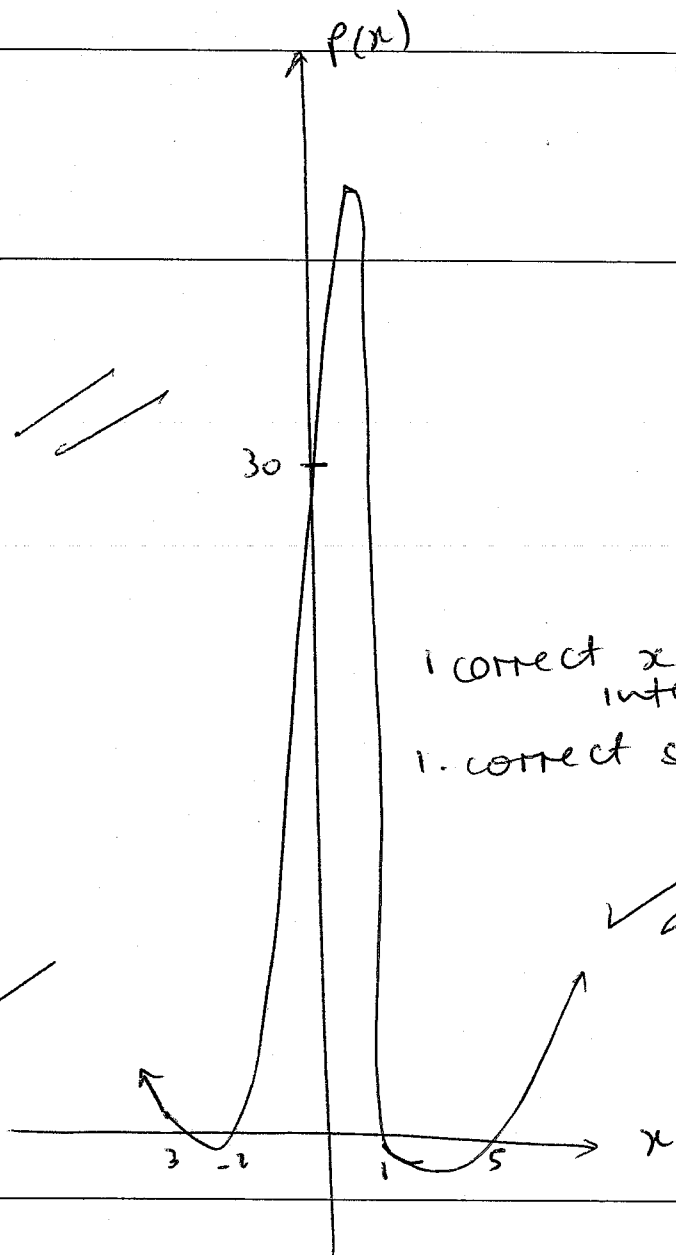
$$P(-2) = 16 + 8 - 76 + 22 + 30 = 0$$

$\therefore (x-1)$ and $(x+2)$ are factors

$$\begin{array}{r} x^2 - 2x - 15 \\ x^2 + x - 2 \overline{) x^4 - x^3 - 19x^2 - 11x + 30} \\ \underline{-(x^4 + x^3 - 2x^2)} \\ -2x^3 - 17x^2 - 11x + 30 \\ \underline{-(-2x^3 - 2x + 4x)} \\ -15x^2 - 15x + 30 \\ \underline{-(-15x^2 - 15x + 30)} \\ 0 \end{array}$$

$$P(x) = (x-1)(x+2)(x^2 - 2x - 15)$$

$$P(x) = (x-1)(x+2)(x-5)(x+3)$$



c. $2x^3 + 0x^2 + 5x - 3 = 0$

i. $\alpha + \beta + \gamma = 0$

ii.

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$\begin{aligned} \alpha^2 + \beta^2 + \gamma^2 &= 0 - 2\left(\frac{5}{2}\right) \\ &= -5 \end{aligned}$$

iii.

$$\begin{aligned} \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\ &= \frac{5}{3} \end{aligned}$$

